# Calendar

Spring 2027

Last modified: 2026-10-30

The topics and dates of future lectures may change.

Spring quarter runs from Tuesday, March 30 to Friday, June 11, 2027, with finals week from Monday, June 7 to Friday, June 11. Class meets on Mondays, Tuesdays, Wednesdays and Fridays, 9:00–9:50am, in CF 420. There is no class on Monday, May 31, for Memorial Day.

Mondays, Tuesdays and Wednesdays are lectures from these notes, apart from exam days and Memorial Day. Fridays are lab sessions, for working on the homework in class, except on the three Fridays that hold a midterm review.

Homework is due at 11:59 pm on its due date, always a Friday. Each self-grade is due one week after that homework's solutions are posted. The self-grade dates below assume the solutions go up at the deadline.

The three midterms are in class:

- Midterm 1, Monday, April 26: weeks 1 to 4, from the course overview through normalization and residual connections.
- Midterm 2, Monday, May 17: convolutional networks, sequence models, language modeling, attention, and Transformers' self-attention and positional encoding.
- Midterm 3, Tuesday, June 1: the lectures from Transformer pretraining through running deep learning experiments.

Each midterm covers the lectures since the one before it, and has an in-class review session on the Friday before the exam.

The final exam is optional. Its day and time are still to be set (TBD): WWU's finals schedule[1] sets each exam's slot from the class's start time and meeting days. The final's review session will be held during finals week, at a time TBD.

| | Monday | Tuesday | Wednesday | Friday |
|---|---|---|---|---|
| **Week 1**<br>Mar 28 – Apr 3 | | Syllabus<br>Course overview | Machine learning review (from lds[2])<br>Tensors (begun) | Lab session: Homework 1 |
| **Week 2**<br>Apr 4 – Apr 10 | Tensors and automatic differentiation | Feedforward networks: layers and activations | Feedforward networks: output units, losses, representational power | Lab session: Homework 1 |

[1] https://registrar.wwu.edu/calendars/finals
[2] https://morrison-lab.github.io/lds/

| | Monday | Tuesday | Wednesday | Friday |
|---|---|---|---|---|
| **Week 3**<br>Apr 11 – Apr 17 | Backpropagation | Optimization: minibatch stochastic gradient descent, momentum | Optimization: Adam, learning-rate schedules and warmup, initialization | Lab session: Homework 2<br>**Homework 1 due** |
| **Week 4**<br>Apr 18 – Apr 24 | Regularization: weight decay, early stopping | Regularization: dropout, data augmentation | Normalization: batch and layer normalization; residual connections | **Midterm 1 review**<br>**Homework 1 self-grade due** |
| **Week 5**<br>Apr 25 – May 1 | **Midterm 1** | Convolutional networks: convolution arithmetic | Convolutional networks: pooling, classic architectures | Lab session: Homework 2 |
| **Week 6**<br>May 2 – May 8 | Convolutional networks: transfer learning | Sequence models: embeddings, recurrent networks | Sequence models: LSTMs and GRUs | Lab session: Homework 3<br>**Homework 2 due** |
| **Week 7**<br>May 9 – May 15 | Language modeling | Attention: encoder-decoder attention | Transformers: self-attention, positional encoding | **Midterm 2 review**<br>**Homework 2 self-grade due** |
| **Week 8**<br>May 16 – May 22 | **Midterm 2** | Transformers: pretraining | Generative models: autoencoders and variational autoencoders | Lab session: Homework 3 and Homework 4<br>**Homework 3 due** |
| **Week 9**<br>May 23 – May 29 | Generative models: generative adversarial networks | Generative models: diffusion | Running deep learning experiments: hyperparameter types, sweeps, compute budgets, reproducibility | **Midterm 3 review**<br>**Homework 4 due**<br>**Homework 3 self-grade due** |
| **Week 10**<br>May 30 – Jun 5 | **No class: Memorial Day** | **Midterm 3** | Applications and current topics: large language models, fine-tuning, speech and vision; ethics | Lab session: Homework 5<br>**Homework 5 due**<br>**Homework 4 self-grade due**<br>**Course evaluations** |
| **Finals week**<br>Jun 7 – Jun 11 | **Optional final exam**<br>Day and time TBD, per the WWU finals schedule[3]<br>Review session: during finals week, time TBD<br>**Homework 5 self-grade due** (Friday, Jun 11) | | | |

[3] https://registrar.wwu.edu/calendars/finals